Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/23/2009
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	chaikin@NYU.edu	**Fax:**	FY 212 995-4016
PI Organization Type:	UNIVERSITY	**Phone:**	212 998-7694
Organization Name:	New York University		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10003	**Congressional District:**	8
Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
Contact Email:	william.v.meyer@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Russel, William (Princeton University) Hollingsworth, Andrew (New York University)		
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Task Description:	This experiment consists of microscopy, microscopic and mesoscopic manipulation and control of colloidal samples to probe the essential features of the hard sphere disorder-order transition and the properties of the ordered solid phase that results. Many samples, in custom-designed microscope slide style cells that permit many different volume fractions and the application of external fields such as temperature gradients, periodic pattern imposition, and force application through the use of optical tweezers, will permit measurements from the disordered fluid phase, through the coexistence region, into the fully crystalline solid, and beyond into a glass-like phase. The dispersions will consist of colloidal hard spheres, in the 0.65 to 1.0 micron diameter range, with controlled polydispersity, consisting of poly (methyl methacrylate) (PMMA) in an index-matched fluid. Microscopy will allow for the observation of the position of individual colloidal particles and thus the determination of local crystalline or liquid structure, defect structure, observation of nucleation and growth of individual crystallites, the spontaneous formation of dendrites, the effect of periodic and other imposed patterns, field induced local melting and freezing, and the dynamics of liquid-crystal, liquid-glass and solid-glass interfaces. The applied fields will allow for controlled nucleation and growth, induced metastable phases, susceptibility, and rheology measurements. A simple design with many 50 microliter samples in a single sample holder on the microscope platen will allow interchange between the many samples to be studied. Static light scattering and confocal microscopy (replacing dynamic light scattering) will be done on the same samples. The static scattering image will be acquired with a CCD for Bragg scattering and structural studies. Confocal microscopy will allow the 3-d study of local affects. We expect to determine the volume fractions at which a fluid changes to a crystal, find the crystal structure of the equilibrium solid phase, contrast nucleation and growth in gravity and microgravity, especially in the region of the ground based glass transition (which we found does not occur in microgravity), quantitatively test the dendritic growth model, explore the role of polydispersity on the phase diagram and the structure of the different phases and explore the susceptibilities to external fields and the microrheology of the different phases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Characterization of crystal formation in the microgravity environment can lead to a greater understanding of how gravity affects many kinds of colloidal materials. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices. The experiment employs a material that assumes multiple phases, which depend only on concentration. The simplest colloidal dispersion consists of hard spheres, which interact through hydrodynamic and Brownian forces, but feel no direct force before an infinite repulsion at contact. The hard sphere phase diagram, including the disorder-order transition, has been well established, but many-body interactions complicate the understanding of the kinetics of the transformations and the mechanical properties of disordered/metastable fluids and the ordered solid. Our experiments on these systems in microgravity, where the effects of sedimentation and convection are absent, have yielded several pleasant surprises. These included crystals (that were glasses on earth) and dendritic growth. In addition, Bragg images during nucleation and growth confirm some anticipated but previously unobserved features of the crystal growth process.
	Ground based support for our flight projects BCAT-4, BCAT-5 The Binary Colloidal Alloy Test (BCAT-4-Poly and BCAT-5-Poly) will be used to study the properties of concentrated systems comprised of small colloidal particles in an index matching fluid. In two samples, all but 1.0 and 0.2% of the spherical colloids are monodisperse "hard spheres"; the (minority) seed particles are about 11.5 times larger in diameter. These inclusions may induce crystallization by heterogeneous nucleation, an effect confounded by earth's gravity. The theoretical prediction is that the use of the right size and concentration of seed particles can be used as a way to control the size of crystallites by reducing the free-energy barrier associated with crystal nucleation. The other BCAT samples consist of polydisperse and monodisperse colloidal glasses. In previous NASA microgravity experiments, we discovered that colloidal suspensions in the volume fraction range 0.58-0.64, which showed glassy behavior but did not crystallize in 1 g, readily formed crystals in microgravity. Although subsequent experiments with near-density-matched samples in normal gravity tended to confirm our results, the samples were far from ideal hard particles due to interactions inherent in the density matching process. Recent numerical simulations suggest that crystal nucleation in suspensions of colloidal spheres with broad size distributions should be suppressed; that is, vitrification at high supersaturations should yield colloidal glasses that are genuinely amorphous. We have extended our earlier microgravity experiments by studying the effects of polydispersity on crystallization in this volume fraction range. The control sample will be as monodispersed as possible, e.g., ~ 5%. Another class of colloidal particles prepared for BCAT experiments includes ellipsoids. We have found that depending on the ellipsoid shape, these particles can pack more densely than spheres in both the crystalline and random configurations. What is particularly interesting is that the difference in packing fraction between random and crystal is smaller for ellipsoids than for spheres; thus, the driving force for crystallization is reduced. Such behavior is reminiscent of the nonergodicity of glasses. In fact, following our work on granular ellipsoids, several groups have numerically tried to find a fluid-crystal transition from computer simulations. As yet there have been no reports of crystallization. While simulations and experiments with granular materials can provide information about packing they do not let us investigate the thermodynamic system the same way that experiments on colloidal systems will. We therefore propose a series of experiments on nonspherical, hard colloidal systems. Preliminary tests will be conducted using a BCAT cell filled with monodisperse hematite-silica (core-shell) ellipsoids. The PMMA colloidal spheres were prepared by dispersion polymerization using a poly(12-hydroxystearic acid)-g-poly(methyl methacrylate) stabilizer prepared in this lab. The 3.8 micron diameter PMMA particles were prepared by a seeded growth process, using PMMA particles prepared by the dispersion polymerization process. The polydisperse sample was prepared with a different batch of PHS-g-PMMA stabilizer that resulted in an unexpected polydispersity (approx. 11%) using twice the normal stabilizer addition (10% versus 5% based on the monomer charge). The scanning electron microscopy (SEM) pictures were taken at the Imaging and Analysis Center located at Princeton University. Small crystallites were observed in the polydisperse sample at effective volume fractions of 0.51 (just above the freezing transition) and 0.55 (just above the melting transition). The refractive index matching fluid utilized a mixture of 55% (w/w) tetrahydronaphthalene (tetralin) and 45% (w/w) decahydronaphthalene (decalin), a mixture of cis and trans isomers. The PMMA particles were thermally equilibrated with the solvent mixture to accelerate the particles swelling process. BCAT sample cells were filled Zin Technologies in Brookpark, OH. The hollow silica ellipsoids were prepared at NYU CSMR by dissolving the hematite cores with 20% hydrochloric acid. These particles have an aspect ratio of about 3:1 and will be refractive index matched with

Task Progress:	dimethylsulfoxide solvent. The synthesis followed the procedure developed by Sacanna et al. (2006). Hematite spindle-like particles stabilized by poly(vinylpyrrolidone) were prepared first. Using tetramethyl ammonium hydroxide, several shells of silica were grown via the polymerization of tetraethoxysilane. Several grams of material were produced. The ellipsoidal particles were intended to be flown on BCAT-5, but solvent compatibility issues prevented this from happening. The dimethylsulfoxide (DMSO) solvent interacted with the Viton O-rings used for secondary confinement in the BCAT sample cell module. However, the Teflon-lined white rubber septa did seal the sample cells without leakage problems. The silica ellipsoids are planned for future microgravity experiments. Two batches of PMMA particles (8.1 g of ADH029 and 9 grams ADH030) were also prepared for Prof. Barbara Frisken (Simon Fraser University) and Dr. Art Bailey. These particles were used in their BCAT-5 sample preparation to study the competition between phase separation and crystallization. The latter batch was characterized with static and dynamic light scattering by Dr. Juan Sabin Fernandez, Simon Fraser University. BCAT-4 Samples 8 and 9: November 9, 2008 ISS observations White light from a flashlight and the camera flash were used to detect the colorful Bragg reflections produced by colloidal crystals formed following the sample mixing. The BCAT-4 experiment was set up, initiated, and optimized by astronauts. The BCAT-4 hardware includes a Slow Growth Sample Module, Magnet and Magnet Keeper (identical in design to the BCAT-3 hardware) and will use the Maintenance Work Area (MWA), DCS760 Camera and EarthKAM software for manual and automated time-lapse photography during six sessions of homogenizing and photographing the 10 samples in microgravity, where the masking effects gravity are removed. Dr. Greg Chamitoff, Flight Engineer and Science Officer on the International Space Station, has been working during his "free time" to get additional photos of the BCAT-4 experiment. Samples 8 and 9 (NYU, Chaikin/Hollingsworth) are the monodisperse and polydisperse "glass" samples, which are designed to study the effects of particle size distribution (PSD) on the phase behavior and crystallization kinetics in dense colloidal fluids. These two samples have significantly different particle size distributions (PSDs): the Sample 9 PSD is symmetrical with a polydispersity of 6.1% which is typical for particles of this size (ave. diameter, unswollen: 0.33 microns). The other sample PSD is negatively skewed with a polydispersity of 10_11% (ave. diameter, unswollen: 0.56 microns). Both samples crystallized after mix-melting. The effective volume fraction of both samples is 0.60, a density at which crystallization is normally not observed under normal (1-g) gravity conditions. The sample 9 observation confirms our previous results aboard the STS-73 Space Shuttle. These images were photographed eight days after the sample homogenization. Greg observed both blue and green when illuminating the samples with a minimag lite, and he has been able to capture both colors with the camera. The sample imaging is difficult due to the acute angle(s) of the flash and camera. The formation of crystals in samples 8 and 9 are significant scientific findings. The crystal nucleation rate is expected to be very slow for small (< 1 micron diameter) and polydisperse (> 5%) colloidal particles at this volume fraction. Previous experiments on colloidal crystallization using this type of suspension show that crystal formation was suppressed when polydispersity exceeded 12%. Our preliminary results in microgravity are intriguing, and with this information we are gaining insight into how size polydispersity may influence the location and existence of a "glass transition". This research affects a broad range of products including medicine, plastics, cosmetics, paints, ceramics and metal/material processing. In December 2008, Mission Specialist Dr. Sandra Magnus, the astronaut performing the current BCAT-4 experiment, observed the signature iridescent colors in our Sample 10, another SeededGrowth experiment. This one contains a higher concentration of spherical seed particles (1%). The high-resolution Kodak camera should provide clear pictures of the seeded heterogeneous nucleation and crystal growth result. The BCAT-4 sample cells represent a significant improvement over the earlier CDOT sample cells that we originally flew on the Space Shuttle.
Bibliography Type:	Description: (Last Updated: 09/29/2025)
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Abstracts for Journals and Proceedings	Hollingsworth AD, Leunissen ME, Irvine WTM, Chaikin PM, van Blaaderen A. "Charged colloids in low polar solvents." Meeting of the American Physical Society, New Orleans, LA, March 10-14, 2008. Bulletin of the American Physical Society, March 2008. , Mar-2008
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